

ANNEXURE - 2 : Personnel Safety

Name of Plant / Unit	Date	Time
Equipment / System		
MOC Number & Description		

Name of PSSR Team Leader	Emp. Code	Designation	Dept.	Sign
Name of PSSR Team Members	Emp. Code	Designation	Dept.	Sign
			Safety / PSM	
			Operations	
			Process	
			Mechanical	
			Inspection	
			Civil	
			Electrical	
			Instrumentation	
			Fire	
			Others	

S. N	Description	Category of Punch Point	Checked by	Remarks
1.	Are following warning signages marked/displayed? No Use of Mobile Phones Restricted Area Authorized entry Emergency Contact Numbers High Noise Area Marking H2S Prone Area Marking High Pressure Marking Area Physical Hazards / Obstruction Hazards Marking Radiological Hazard marking Escape route & Emergency exit Safety shower Assembly point Nitrogen & Utility station		Operation	

Procedure for Pre Start-up Safety Review

2.	Has adequate and appropriate personnel protective equipment((PVC gloves, chemical splash proof goggles, ear plugs/muffs, face shield, PVC apron, and escape masks, Flame proof torch, Escape B.A Set available at the work locations?)		Operation	
3.	Has adequate and appropriate meter(LEL / Oxygen / H2S / CO/SO2/CL/Hydrogen)available?		Operation	
4.	Is the required number of wind sock installed and all of them in working condition & visible?		Operation	
5.	Are safety shower installed as per requirements Is the shower colour coding done as per procedure?		Operation	
6.	Has adequate and appropriate no. of KIT (like Chlorine cylinder / tonner or other toxic gas cylinder leakage arresting kits,pill control kit) available?		Operation	
7.	Whether vessel / equipment skirt openings are covered?		Operation	
8.	Has all material handling operations have Operating instructions, Special Instructions, Do's and Don'ts -posted nearby?		Operation	
9.	Are the operating personnel aware of the chemical risks? Are the chemical safety details displayed in brief? MSDS available and displayed?		Operation	
10.	Training and competency records of employees & contractors:		Operation	
11.	HSEF Training (OISD, ERDMP,first aid,fire fighting, first work permit system, H2S awareness etc. .) conducted and record available?	A	Operation	
12.	Functional Training conducted and record available?	A	Operation	
13.	Are all communication system like Walkie talkie,Field phone,PA system,MCP are healthy?	A	Operation	
14.	Noise level survey done & made report? Are the noise levels below the acceptable values? First aid box available ?		Operation	
15.	Survey of moving/rotating machinery done? Are adequate guards provided for protection against: Rotating parts/pinch/nip points/belts/cutting edges/hot / cold surface?		Mechanical	
16.	Whether illumination survey done? Whether illuminatin level is above standard values?		Electrical	
17.	Are all aviation lights working?	A	Electrical	
18.	Are personnel protection provided against human contacts with hot surfaces which may reach or exceed 55° C?		All	
19.	Are the Floor,platforms / stairs survey done? guard rails i.e. top, mid & toe guard provided? & free from obstructions, slip and trip hazards?	A	All	
20.	Other punch points		All	

ANNEXURE - 4 : Fire Protection

Name of Plant / Unit	Date	Time
Equipment / System		
MOC Number & Description		

Name of PSSR Team Leader	Emp. Code	Designation	Dept.	Sign
Name of PSSR Team Members	Emp. Code	Designation	Dept.	Sign
			Safety / PSM	
			Operations	
			Process	
			Mechanical	
			Inspection	
			Civil	
			Electrical	
			Instrumentation	
			Fire	
			Others	

S.N	Description	Category of Punch Point	Checked by	Remarks
Document Check Points				
1.	Have emergency response plans and scenarios been updated to reflect the new facilities, and available at both the new/modified installation and at the fire and emergency response centres/Control Room ?	A	Fire	
2.	Are the inert gases extinguishing systems checked for the following? a) Solenoid -'functional check' records 'Inert Gas cylinders pressure checked -main reserve. b) Equipment's suitable for area classification	A	Instrument/ Fire	

Refinery Integrated Management System

Procedure for Pre Start-up Safety Review

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	c) Functional check records d) Pressure gauges calibration records e) Air loop/water loop leak/pressure-tested ok			
• Field Check Points				
1.	Is rescue plan for vulnerable locations available?		Fire	
2.	Are there nos. of Hydrants, Monitors, Water spray deluge system, Foam trolley as per the design?	A	Fire	
3.	Is there Foam fixed semi fixed mobile system required and available?		Fire	
4.	Are there Gaseous extinguishing system available as per design?	A	Fire	
5.	Are there Rim seal foam package system available as per design??		Fire	
6.	Do the firewater monitors checked for the following? a) Elevation and Traverse movement b) For unrestricted movement c) For ease of operation d) Nozzle checks for Jet Fog Shut-off e) Throw checked	A	Fire	
7.	Is the adequate no. of SCBA Sets installed in field and charged with BA at appropriate pressure and in working condition?	A	Fire	
8.	Fire engine trial run from fire station to plant taken and it is found satisfactory.		Fire	
9.	Portable Fire Extinguishers			
9.1	Are there Portable fire extinguishers (nos., capacities & types) provided according to fire loads?	A	Fire	
9.2	Portable fire extinguishers master list available with tag no., type capacities exact location?		Fire	
9.3	Fire extinguisher locations are visible & accessible?	A	Fire	
9.4	Are locations painted with zebra marking for their easy identification?		Fire	
9.5	Are Extinguishers tagged & instruction labels available?		Fire	
10.	Operability of hydrant checked.	A	Fire	
11.	Fire hoses, hose box, nozzles, available at all Identified locations.	A	Fire	

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Procedure for Pre Start-up Safety Review

12.	Sprinkler/Deluge valve operation checked and found ok.	A	Fire	
13.	All Fire water piping network checked and is as per drawings/P&ID/Isometrics.	A	Process / Fire	
14.	Are the fire water pumps checked for following?			
14.1	System installation as per P & IDs	A	Process	
14.2	Mechanical completion records of fire system	A	Fire	
14.3	Fire water pumps /engines	A	Fire	
14.4	Batteries in condition	A	Fire	
14.5	Diesel tanks full	A	Fire	
14.6	Auto start system for fire pump OK	A	Fire	
14.7	Functionally checked & found OK	A	Fire	
14.8	Jockey pump operating parameters maintained		Fire	
14.9	Access road(s) clear for fire tender from Fire station to operating areas?	A	Fire	
15.	Approach to hydrants / extinguishers clear and accessible?	A	Operation	
16.	Painting of fire water piping and installations complete in all aspects		Operation	
18.	Other punch points		All	

ANNEXURE - 5 : F & G and other Communication Systems

Name of Plant / Unit	Date	Time
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Name of PSSR Team Leader	Emp. Code	Designation	Dept.	Sign
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			Safety / PSM	
			Operations	
			Process	
			Mechanical	
			Inspection	
			Civil	
			Electrical	
			Instrumentation	
			Fire	
			Others	

S.N	Description	Punch point category	Checked by	Remarks
Document Check Points				
1.	PA System checked for a) Master list of sets available? b) Functional check records available?	A	Electrical	
2.	Wireless set – a) Master list of sets available? b) Integrity & legal requirement fulfilled c) Functional check records available?	A	Electrical	
3.	Telephones system checked for a) Flame proof field phone integrity? b) Master list of sets available? c) Functional check records available?	A	IT/Electrical	

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Procedure for Pre Start-up Safety Review

• Field Check Points

1.	Locations of all MCPs checked and found OK. List of MCP displayed in the unit?	A	Fire/Operation	
2.	Beacon & Siren system (if installed) working and audible/Visible at all places?	A	Fire & Electrical	
3.	Do the F G system, emergency alarm, beacons and communication systems exist as per the requirement?	A	Fire/Instrument	
4.	Location of Gas detectors, Smoke detectors, Quartz bulbs, Toxic gas detectors, UV detectors, Radioactive detector as per drawing?	A	Instrument	
5.	F & G system calibration and actuation checked? Settings as per recommendations? Are there records available?	A	Instrument	
6.	Actuation of DV, Inergen release, perimeter lights etc. with actuation of Flammable, Smoke, Toxic gas detectors checked? Are there records available?		Instrument	
7.	System installation as per P & IDs - Loop connectivity to DCS & fire panel checks? Records available?	A	Instrument	
8.	Alarms at local control room and fire control room are functioning?	A	Instrument	
9.	Hotlines system checked for Functional check, records available?	A	IT/Operation	
10	Other punch points		All	